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January 2024

A culture of constraints: How head coach turnovers affect the backroom staff and player development in professional football. International Journal of Sports Science & Coaching, Ahead of Print: 1-10, 2023.

A randomized, double-blind, placebo-controlled, repeated-dose pilot study of the safety, tolerability, and preliminary effects of a cannabidiol (CBD) and cannabigerol (CBG) based beverage powder to support recovery from delayed onset muscle soreness (DOMS). Journal of the International Society of Sports Nutrition, Ahead of Print: 1-18, 2023.

A systematic review and meta-analysis of wearable satellite system technology for linear sprint profiling: Technological innovations and practical applications. Journal of Strength and Conditioning Research, Ahead of Print: 1-14, 2023.

Analytics, have some humility: A statistical view of fourth-down decision making. arXiv:2311.03490: 1-55, 2023.

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Beyond the Scoreboard: Redefining performance staff assessment in elite sports organizations. Sport Performance & Science Reports, November: 210 v1, 2023.

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Changes in head staff members in male elite-level football teams are associated with increased hamstring injury burden for that season: The UEFA Elite Club Injury Study. BMJ Open Sport & Exercise Medicine, 9:e001640, 2023.

Current implementation and barriers to using blood flow restriction training: Insights from a survey of allied health practitioners. Journal of Strength and Conditioning Research, Ahead of Print: 1-10, 2023.

Discriminant function analysis reveals which combination of measures from the NFL scouting combine predict NFL performance. Journal of Expertise, Vol. 6(2):130-163, 2023.

Effects of an individualized vs standardized Vitamin D supplementation on the 25(OH)D level in athletes. Nutrients, 15: 4747, 2023.

Explaining variation in sweat sodium concentration: Effect of individual characteristics and exercise, environmental, and dietary factors. Journal of Applied Physiology, 133: 1250-1259, 2022.

Exploring the interplay between metabolic power and equivalent distance in training games and official matches in soccer: A machine learning approach. Frontiers in Physiology, 14:1230912, 2023.

Explosive in not a term defined in the International System of Units and should not be used to describe neuromuscular performance. International Journal of Strength and Conditioning, Vol. 3(1): 1-10, 2023.

Fifth metatarsal strain distribution during cutting motions in soccer. Sports Biomechanics, Ahead of Print: 1-18, 2023.

From data to action: A scoping review of wearable technologies and biomechanical assessments informing injury prevention strategies in sport. BMC Sports Science, Medicine and Rehabilitation, 15: 169, 2023.

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Fundamentals or icing on top of the cake? A narrative review of recovery strategies and devices for athletes. Sports, 11: 213, 2023.

Hamstring and knee injuries are associated with isometric hip and trunk muscle strength in elite Australian Rules and Rugby League players. Journal of Science and Medicine in Sport, Ahead of Print: 1-25, 2023.

Inertial measurement unit analysis for providing greater diagnostic value during the modified 5-0-5 change of direction test. The Journal of Sport and Exercise Science, Vol. 7(2): 54-61, 2023.

Measurement properties of upper extremity physical performance tests in athletes: A systematic review. Brazilian Journal of Physical Therapy, Ahead of Print: 1-18, 2023.

Multiple ankle injuries are associated with an increased risk of subsequent concussion in National Football League players. Foot & Ankle Orthpaedics, Vol. 8(4): 1-6, 2023.

Napping behavior, daytime sleepiness, and arousal in high performance athletes and non-athlete controls. Journal of Sports Sciences, Ahead of Print: 1-9, 2023.

Nighttime smartphone use, sleep quality, and mental health: Investigating a complex relationship. Sleep, 46: 1-12, 2023.

Open fractures in National Football League athletes: Analyzing performance and return to sport. Spartan Medical Research Journal, 8(1):87846, 2023.

Potential health risks of foodborne performance-enhancing drugs in competitive sports. Heliyon, 9:321104, 2023.

Quantifying demands on the hamstrings during high-speed running: A systematic review and meta-analysis. Scandinavian Journal of Medicine and Science in Sports, Ahead of Print: 1-21, 2023.

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Review of media representation of sport concussion and implications for youth sports. Clinics in Sports Medicine, Ahead of Print: 1-14, 2023.

Sciatic nerve compression after a chronic proximal hamstring tear: A report of two cases and a narrative review of the literature. Life, 13:1762, 2023.

The effects of knee ligament load using simulated hip abductor and hamstring muscle strengthening during cutting maneuver. Medicine, 102:46, 2023.

The effects of relaxation techniques following acute, high intensity football training on parasympathetic reactivation. Frontiers in Sports and Active Living, 5:1267631, 2023.

The Molecular Athlete: Exercise physiology from mechanisms to medals. Physiological Reviews, 103: 1693-1787, 2023.

To train or not to train (on match day)? Influence of a priming session on match performance in competitive elite-level soccer. Journal of Sports Sciences, Ahead of Print: 1-9, 2023.

Turf toe injuries in the athlete: An updated review of treatment options, rehabilitation protocols, and return to play outcomes. Current Reviews in Musculoskeletal Medicine, Ahead of Print: 1-12, 2023.

Where to next for hamstrings? A biomechanical and anatomical perspective. British Journal of Sports Medicine, Ahead of Print: 1-2, 2023.

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A culture of constraints: How head coach turnovers affect the backroom staff and player development in professional football. International Journal of Sports Science & Coaching, Ahead of Print: 1-10, 2023.

Abstract - In professional football, head coach turnovers are arbitrarily perceived as a shortcut solution to convert losing streaks into winning results within time-limited competitive seasons. Despite a speculative mentality guiding turnover decisions, a new coach is invariably appointed with the expectation of improving performance. As often overlooked, however, coaching replacements inevitably create unintended club-wide spillover effects. This study addresses two related club-wide research questions: (1) How do head coach turnovers affect player development at the team level?; and, (2) What are the spillover effects for backroom staff? Undertaken in Brazil, where turnover rates exceed those in Europe, 30 backroom staff practitioners (i.e., assistant coaches, goalkeeping coaches, fitness coaches, physical therapists, physiologists, and a medical doctor) completed semistructured interviews focused on their experiences. A reflexive thematic analysis revealed an overarching culture of constraints driven by conflicting behaviors and working practices. Three themes were highlighted at the team level: pragmatic playing style, methodological mismatch, and delaying team cohesion. At the individual level, four themes were identified: internal disruption, repetitive adaptations, job insecurity, as well as mental and emotional unrest. The findings indicate how within-season coaching turnovers represent a hazard that unintentionally constrains the mutual collaboration among the stakeholders directly involved with player development.

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A randomized, double-blind, placebo-controlled, repeated-dose pilot study of the safety, tolerability, and preliminary effects of a cannabidiol (CBD) and cannabigerol (CBG) based beverage powder to support recovery from delayed onset muscle soreness (DOMS). Journal of the International Society of Sports Nutrition, Ahead of Print: 1-18, 2023.

Background: Cannabinoid-containing products are marketed to athletes as promoting recovery, in spite of a lack of data on their safety and effects. This randomized, double-blind, placebo-controlled, repeated-dose pilot study tested the safety, tolerability, and preliminary effects on recovery of a formulation containing cannabidiol (CBD; 35 mg), cannabigerol (CBG; 50 mg), beta caryophyllene (BCP; 25 mg), branched-chain amino acids (BCAAs; 3.8 g), and magnesium citrate (420 mg).

Methods: Exercise-trained individuals (N = 40) underwent an experimental induction of delayed onset muscle soreness (DOMS) and completed follow-up visits 24-, 48-, and 72-hours post-DOMS. Participants were randomized to active or placebo formulation, and consumed the formulation twice per day for 3.5 days.

Results: There was one adverse event (AE) in the active group (diarrhea) and two AEs in placebo (dry mouth; eye rash/swollen eye). There was 100% self-reported compliance with formulation consumption across the two groups. For the primary outcome of interest, the estimate of effect for ratings of average soreness/ discomfort 72 hours post-DOMS between active and placebo groups was −1.33 (85% confidence interval = -2.55, −0.10), suggesting moderate evidence of a treatment difference. The estimate of effect for the outcome of ratings of interference of soreness, discomfort, or stiffness on daily activities at work or home 48 hours post-DOMS was −1.82 (95% confidence interval = -3.64, −0.01), indicating a treatment difference of potential clinical importance. There was no significant effect between active and placebo groups on objective measures of recovery, sleep quality, or mood disturbance.

Conclusions: The tested formulation reduced interference of DOMS on daily activities, demonstrating its improvement on a functional aspect of recovery.

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A systematic review and meta-analysis of wearable satellite system technology for linear sprint profiling: Technological innovations and practical applications. Journal of Strength and Conditioning Research, Ahead of Print: 1-14, 2023.

Abstract - An emerging and promising practice is the use of global navigation satellite system (GNSS) technology to profile team-sports athletes in training and competition. Therefore, the purpose of this narrative systematic review with meta-analysis was to evaluate the literature regarding satellite system sensor usage for sprint modeling and to consolidate the findings to evaluate its validity and reliability. Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, an electronic search of the databases, PubMed and SPORTDiscus (EBSCO), was conducted. Concurrent validity and reliability studies were considered, and 16 studies were retained for the review from the initial 1,485 studies identified. The effects on outcomes were expressed as standardized mean differences (SMDs, Cohen's d) for each outcome (i.e., maximal sprint speed [MSS], the acceleration constant [τ], maximal theoretical velocity [V0], relative force [F0], and relative power [Pmax]). Effect magnitudes represented the SMD between GNSS-derived and criterion-derived (i.e., radar and laser) and resulted in the following estimates: small for MSS (d = 0.22, 95% CI 0.01 to 0.42), τ (d = -0.18, 95% CI -0.60 to 0.23), V0 (d = 0.14, 95% CI -0.08 to 0.36), relative F0 (d = 0.15, 95% CI -0.25 to 0.55), and relative Pmax (d = 0.21, 95% CI -0.16 to 0.58). No publication bias was identified in meta-analyzed studies and moderator analysis revealed that several factors (sampling rate and sensor manufacturer) influenced the results. Heterogeneity between studies was considered moderate to high. This highlighted the differences between studies in sensor technology differences (i.e., sampling rate, sensor fusion, and satellite network acquisition), processing techniques, criterion technology used, sprint protocols, outcome reporting, and athlete characteristics. These findings may be useful in guiding improvements in sprint modeling using GNSS technology and enable more direct comparisons in future research. Implementation of all-out linear sprint efforts with GNSS technology can be integrated into sport-specific sessions for sprint modeling when robust and consistent data processing protocols are performed, which has important implications for fatigue monitoring, program design, systematic testing, and rehabilitation in individual and team sports.

Practical Applications - Although there are some methodological challenges to sprint modeling using GNSS technology, validity and reliability studies suggest that sprint modeling can be performed and show promise for sprint monitoring in the field. The key consideration is that the most up-to-date

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modeling practices should be used consistently with similar sprint protocols because the methodology will affect sprint profiling outcomes. As suggested in previous research, the integration of maximal standardized linear sprint efforts can be integrated into training sessions with team-sports and in dividual sport athletes. Systematic sprint modeling could allow performance teams to monitor fatigue and readiness, better-informing periodization and training stimulus prescription. This has important implications for concepts like “invisible monitoring” (i.e., testing without testing), systematic testing, fatigue monitoring, rehabilitation, performance, and training design.

Analytics, have some humility: A statistical view of fourth-down decision making. arXiv:2311.03490: 1-55, 2023.

Abstract - Expected points (EP) and win probability (WP) are value functions fundamental to strategic in-game decision making in American football, particularly for fourth down decision making. The EP and WP functions which are widely used today are statistical models fit from historical data. These models, however, are subject to serious statistical flaws: selection bias, overfitting, ignoring autocorrelation, and ignoring uncertainty quantification. We develop a machine learning framework that accounts for these issues and extracts our analysis into a decision-making inference. Along the way, we introduce a novel methodological approach to mitigate overfitting in machine learning models. Specifically, we extend the catalytic prior, initially developed in the context of linear models, to smooth our tree machine learning models. Our final product is a major advance in fourth-down strategic decision making: far fewer fourth-down decisions are as obvious as analysts claim.

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Athlete mental health help-seeing: A systematic review and meta-analysis of rates, barriers and facilitators. Psychology of Sport & Exercise, Ahead of Print: 1-48, 2023.

Abstract - Athletes are vulnerable to a range of mental health symptoms, in part due to stressors within the sport environment. An early intervention framework suggests the benefits of routine screening and referral for mental health, however, greater understanding around athlete help-seeking is needed to support referral uptake. This review examined rates of formal help-seeking behavior as well as barriers and facilitators to help-seeking in sport settings. Relevant studies were retrieved from SportDiscus, PubMed and PsycInfo, with unpublished studies identified through contacting authors. Help-seeking rates were meta-analyzed and barriers and facilitators were meta-synthesized. Twenty-two studies were included. Help-seeking rates were reported in 11 studies (N=3415) and the pooled proportion of help-seeking was 22.4% (95% CI 16.2 – 30.2, I2=95.7%). Barriers were reported in 13 studies and facilitators in six, highlighting a range of sporting-specific factors, such as stigma in relation to athlete identity and sport culture, fear of deselection, and concerns around confidentiality in sport settings, in addition to lack of awareness, low mental health literacy, and negative attitudes to services. Normalizing experiences of mental health in sport settings, including through role models, was a key facilitator to help-seeking. Results provide implications for sport organizations to promote help-seeking and athlete mental health, such as through the use of role models, ensuring clarity around confidentiality, stigma reduction interventions, and fostering team cultures that promote mental health. Findings also support the value of sport staff in facilitating help-seeking, and organizational culture changes to foster wellbeing.

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Attention-deficit/hyperactivity disorder-related self-reported symptoms are associated with elevated concussion symptomatology. Journal of Sport & Exercise Psychology, 44: 116-126, 2022.

Abstract - The aim of the present investigation was to provide insight into how postconcussion symptomatology may be altered in individuals exhibiting attention-deficit/hyperactivity disorder (ADHD)–related behaviors and examine factors that may be responsible for driving such relationships. A total of 99 individuals were assessed during the subacute phase of concussion recovery. Inattentive symptomatology, but not diagnosis of ADHD, was related to greater concussion-symptom severity and overall symptoms endorsed. Cluster and factor analyses highlighted that the relationship between ADHD symptomatology and concussion symptomatology was not a function of overlapping constructs being assessed (i.e., concussion-related symptomatology was not a proxy of ADHD-related symptomatology). These relationships were not mediated by parental observations of impairments in behaviors associated with executive functioning (i.e., executive dysfunction was not driving the greater concussion-related symptomatology associated with ADHD-related symptomatology). These findings highlight the importance of moving beyond categorical frameworks of ADHD to, instead, consider the continuum of underlying behaviors.

Beyond the Scoreboard: Redefining performance staff assessment in elite sports organizations. Sport Performance & Science Reports, November: 210 v1, 2023.

Headline: In corporate high-performance settings, Key Performance Indicators (KPIs) play a pivotal role in assessing employee performance, offering several advantages to organizations. KPIs serve as metrics signifying a company's overall health and progress (Schein 2004, Pohl 2022). They promote transparency by measuring efficiency and output, providing insights into business success, whether at the company, departmental, or individual level. Well-defined KPIs set achievable goals, both for individuals and the organization as a whole, enabling performance evaluation to identify high-performers and areas requiring improvement. Additionally, KPIs allow leadership to adapt expectations based on trends, fostering motivation among employees through clear, attainable objectives. Importantly, employees' ability to meet their KPIs can impact bonuses and job security. Furthermore, transparent objectives

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drive alignment and integration across individuals, teams, and departments, facilitating efficient workflows and processes while propelling organizations toward critical outcomes and process goals. This integrated approach enhances communication and collaboration, ensuring that everyone works cohesively toward shared objectives and promoting overall organizational success. In the realm of elite sports, particularly professional football (soccer), assessing staff performance is a complex task, quite distinct from the corporate landscape. Notably, KPIs are scarce in this arena, sparking a debate surrounding the root causes. On one side of the argument, the absence of KPIs may be attributed to a lack of clear objectives within and across disciplines, hindering the establishment of accountability throughout the entire process. Conversely, an alternative reason may be related to the fact that in professional sports, especially team sports, setting up KPIs presents a distinctive challenge due to the intricate nature of the game. Quantifying the impact of support staff on team victories and individual player well-being proves exceedingly challenging due to the multitude of genetic and physiological factors at play. In this context, a recent study explored the correlation between the level of advancement in sports science practices within clubs, as measured by Performance Science Index scores (Buchheit 2022), and the overall performance of clubs, represented by Elo ranking. Surprisingly, the research revealed that advanced sports science and medical practices did not necessarily correlate with on-field performance. This finding suggests that what occurs behind the scenes can sometimes be dissociated from the outcomes observed on the pitch. Consequently, support staff often find themselves solely evaluated based on team outcomes, even when a team performs well despite potential coaching deficiencies. This focus on out comes rather than underlying processes blurs the definition of performance for support staff. Assessing practitioners across various disciplines proves inconsistent, as evaluations often do not consider their domain-specific interventions and their resulting effects. Consequently, practitioners' advancement or departure from the sports industry tends to hinge more on their interpersonal skills and their capacity to navigate the environment rather than their technical proficiency. This challenge extends to evaluating the quality and significance of work at both departmental and individual levels. Additionally, it raises questions about how to approach the interdependent aspects of performance support teams' work, where some KPIs and objectives may be co-owned, shared, or entail dependence between interacting elements. Complicating matters further is the tendency to conflate KPIs for sports like objectives for players to achieve specific fitness levels or in-game KPIs (Cardinale 2022) with the current project KPIs used for staff evaluation. When it comes to athlete KPIs, the complexity deepens, as physically-derived KPIs or benchmarks often show poor correlations with success in team sports (Clubb 2023). This discrepancy arises from the fact that success

in team sports, particularly soccer, goes beyond physical performance and involves scoring goals and overall game strategy. The inability to define KPIs for the sport and the confusion surrounding the types of KPIs needed further compound the complexities faced in this unique context.

Key Findings:

Evaluation methods and KPIs use among practitioners

• The survey conducted among elite sports practitioners revealed essential insights into the use and structuring of KPIs in sports organizations, and offered valuable guidance for the design of department and staff-specific evaluation KPIs.

• Present findings collectively reflect the nuanced landscape of KPI usage and the critical role they play in enhancing sports organizations' performance evaluation processes.

• The two main methods used to assess staff performance include: ◦ Objective Performance Metrics and KPIs ◦ Subjective Assessment and Feedback

• These evaluations happen via regular reviews and meetings; technology and data-driven tools are often integrated into these methods to support the assessment process.

• Among the 51 practitioners interviewed, 87% reported setting objectives for Practitioners/Departments.

• The main reason that motivates the use of KPIs is with no surprise to measure the impact of practitioners and departments (i.e., the MDT).

• The large majority of reported KPIs were both department and staff-specific. • The main barriers to KPIs implementation are that they are not easy to define (60%) and the lack of support from superiors and key stakeholders (26%).

• When evaluating staff performance, 85% of the practitioners believe that the evaluation should be more process- than outcome-driven (i.e., 70-80% for process-driven being the preferred rating).

Department (MDT) KPIs

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• Practitioners held the highest preference for MDT overall KPIs related to "Overall player availability" with 66% very high relevance, followed by "Quality of communication with technical staff about players' health and performance" at 60%.

• Injuries and rehabilitation KPIs were highly valued by practitioners, with the two most preferred KPIs being "The overall collaboration process with the coaching staff for player reintegration" (75%) and "The quality of diagnostics" (67%).

• The KPI with the highest preference among practitioners in relation to communication was "Clearly established communication lines and responsibilities between all the staff," with 71% indicating very high relevance.

• The most preferred player testing, monitoring, and bench marking KPIs among practitioners were "Creating individual player profiles with practical information" (60%), followed by "Performance analysis using match data" (52%).

• Practitioners indicated that the most highly preferred player health and fitness monitoring KPIs, were "Offering individual development plans for players regarding endurance, strength, speed, and movement", as well as "Offering individual injury prevention routines following screenings", both receiving 58% relevance.

• Among the Planning and Periodization KPIs, the highest preference was for the "2-6 weeks team periodization plan for team training" (48%).

• Practitioners express polarized opinions in terms of responsibility regarding RTP duration; 55% think that "It is the nature of the injury, human physiology, and the context" (55%) while 30% believe that it's "≥70% MDT" responsibility.

• For squad and player availability, practitioners leaned towards collective responsibility, with the most preferred option being "50% MDT/50% coaching staff" (43%).

• The KPI suggestions with the highest preference among practitioners for building a sustainable and evolving long-term club methodology and approach were "Sustainability of practices over time" with 65% very high relevance, and "Complete and detailed documentation available of

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screening/testing/team training/locomotor (GPS) load/individual strength training throughout the season" with 60% very high relevance.

Individual KPIs

• The personal traits KPIs with the two highest preferences were "Willingness to progress and grow personally" at 75% and "Strong inter-personal skills - build and maintain relationships" at 71% in terms of very high relevance.

• When assessing individual communication and soft skills KPIs, the two highest-preference indicators among practitioners were the "Ability to provide and receive critical and productive feedback to/from co-workers in the department" at 73% and "Quality of global communication with colleagues in the same discipline" at 69%.

• Among the Job Delivery KPIs, the two with the highest preference among practitioners were "Deliver best-practices sessions/pieces of advice/treatment" with 56% very high relevance and "Job always done timely in relation to team practice at the sport pace" with 54% very high relevance.

• Organizational traits KPIs received mixed consensus and relatively lower ratings in the "very high relevance" category, with "Dynamics of rest/leave day" being rated as highly relevant by 48% of the practitioners, and "Global investment" receiving a very high relevance rating from 31% of respondents.

Causes of sleep deprivation in competitive athletes: A scoping review. Science & Sports, Ahead of Print: 1-8, 2023.

Purpose: Inadequate sleep has a negative impact on the psychological and physiological health of elite athletes. Studies have shown a correlation exists between sleep deprivation and a drop in performance. Despite rising interest in sleep science, current research focus is on outcome and the cause of sleep deprivation is a secondary consideration. This scoping review aims to identify themes in peer-reviewed studies that are related to the possible causation of sleep deprivation in competitive athletes.

Method: A scoping review was conducted using the PRISMA-ScR guideline. Three databases were used to research for the literature PubMed, Medline and EBSCO.

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Results: After applying inclusion and exclusion criteria 16 peer-reviewed studies were identified. The studies were categorized into two sub-categories of short- and long-term causes of sleep deprivation. Five themes were identified: travel, night performance, prior, during and following competition and concussion.

Conclusion: The findings of this scoping review identified a number a short and long-term causes of sleep deprivation. The causes of the short-term causes can be addressed by implementing some simple interventions such as napping, relaxation techniques and sleep hygiene could aid an athlete performance. However, the long-term causes were linked to concussion, any intervention would need to be studied in greater detail and is beyond the remit of this scoping review.

Changes in head staff members in male elite-level football teams are associated with increased hamstring injury burden for that season: The UEFA Elite Club Injury Study. BMJ Open Sport & Exercise Medicine, 9:e001640, 2023.

Objective: To evaluate whether a change of head coach or other head staff before or during a season is correlated to hamstring injury (HI) burden in male elite-level football (soccer) in Europe.

Methods: The survey was conducted using a questionnaire reporting any staff change within the team. Data about the head staff changes and hamstring injury burdens were collected from 14 teams participating in the Elite Club Injury Study (ECIS) during the 2019/2020, 2020/2021 and 2021/2022 seasons.

Results: On average, replacing the head coach before or during a season happens in every second season. All changes, except for the change of the head coach during a season, indicate an association with an increase in HI burden (ranging from 10% to 81%). However, only changes in the fitness coach and team doctor roles reached statistical significance. The HI burden seems to be influenced by adding new staff members, such as the head of fitness/performance coach in 36% of the teams and the team doctor in 17%. New head coaches starting the season with their own, for the team new, fitness/performance coach was highly associated with increased HI burden ($p<0.001$).

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Conclusions: Bringing their own fitness/performance coaches is common for managers entering a new elite male football club. However, this paper has highlighted that this trend seems to lead to a three-times increase in HI burden. Similarly, replacing the team doctor was also associated with increased HI burden. Instability among head staff members in male elite-level football teams seems associated with increased HI burden during the season.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Coaching replacement is a frequent phenomenon in elite-level football, and its impact on a team's success is debatable.

⇒ Hamstring injury (HI) is the most common injury in professional football (soccer).

⇒ Recent research has indicated that coach replacement might increase the number of muscle injuries within teams.

WHAT THIS STUDY ADDS

⇒ The head coach replacement before or during a season was the most common change among head staff members; it took place in every second team season.

⇒ Replacements of the head of fitness/performance coach and the team doctor were associated with increased HI burden, but an isolated head coach replacement was not.

⇒ The most distinct increase in HI burden was noticed when a new coach simultaneously brought his own new fitness/performance coach to the team.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Elite football clubs must develop and invest in their performance staff to create continuity within their department.

⇒ The transition to a new fitness/performance coach or team doctor should be completed with the utmost care and accompanied by a thorough and precise handover with the former staff.

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Current implementation and barriers to using blood flow restriction training: Insights from a survey of allied health practitioners. Journal of Strength and Conditioning Research, Ahead of Print: 1-10, 2023.

Abstract - This study investigated the use of blood flow restriction (BFR) exercise by practitioners working specifically with clinical or older populations, and the barriers preventing some practitioners from prescribing BFR. An online survey was disseminated globally to allied health practitioners, with data from 397 responders included in analyses. Responders who had prescribed BFR exercise (n = 308) completed questions about how they implement this technique. Those who had not prescribed BFR exercise (n = 89) provided information on barriers to using this technique, and a subset of these responders (n = 22) completed a follow-up survey to investigate how these barriers could be alleviated. Most practitioners prescribe BFR exercise for musculoskeletal rehabilitation clients (91.6%), with the BFR cuff pressure typically relative to arterial occlusion pressure (81.1%) and implemented with resistance (96.8%) or aerobic exercise (42.9%). Most practitioners screen for contraindications (68.2%), although minor side effects, including muscle soreness (65.8%), are common. The main barriers preventing some practitioners from using BFR are lack of equipment (60.2%), insufficient education (55.7%), and safety concerns (31.8%). Suggestions to alleviate these barriers included developing educational resources about the safe application and benefits of BFR exercise (n = 20) that are affordable (n = 3) and convenient (n = 4). These results indicate that BFR prescription for clinical and older cohorts mainly conforms with current guidelines, which is important considering the potentially increased risk for adverse events in these cohorts. However, barriers still prevent broader utility of BFR training, although some may be alleviated through well-developed educational offerings to train practitioners in using BFR exercise.

Practical Applications - Allied health practitioners prescribing BFR exercise for clinical and older clients often screen for contraindications before using this technique, aligning with current evidence-based guidelines (22). Nevertheless, minor side effects to BFR exercise were still reported with greater frequency than in previous surveys, likely because of the specific client demographic targeted in this research. To reduce the likelihood of side effects, practitioners should gradually implement BFR exercise into a client's training program by beginning with lower cuff pressures and exercise volumes and increasing to the desired per centage of AOP and external demands over several sessions. Although barriers prevent some practitioners from pre scribing BFR exercise, some of these could be alleviated by

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well-developed BFR educational offerings. Education providers can use online or face-to-face training, focused on contraindications and screening, determination of BFR pressure, cuff application, and the safety of BFR.

Discriminant function analysis reveals which combination of measures from the NFL scouting combine predict NFL performance. Journal of Expertise, Vol. 6(2):130-163, 2023.

Abstract - The National Football League (NFL) Scouting Combine (the Combine), is an annual track-meet style event where NFL scouts and general managers evaluate newly-eligible players before the upcoming draft. During the Combine, players' height, weight, speed, agility, acceleration, jumping, explosive movement, and strength are measured through their participation in multiple drills such as the 40-yard dash, vertical jump, and bench press. Although numerous studies have tested which individual drills predict NFL success for different positions, the results are often inconsistent across studies. These previous studies rely largely on individual predictors without considering how the various abilities measured at the Combine work together to predict performance. In this study, using discriminate function analysis, we analyze 20 years of data to identify the best combination of skills necessary to achieve varying levels of success for each player position. To date, this study represents the largest, most comprehensive study on the topic. We found that for offensive positions, single measures were often the best predictors of success. By contrast, for defensive positions, we found significant discriminant functions identifying unique combinations of traits that predicted success. We examined success using multiple benchmarks: draft status, number of games played, number of games started, and honors received. All significance tests were two-tailed, alpha = .05. These results indicate that, at least for some NFL positions, scouts and general managers should consider relative performance across multiple drills. Differences between the predictors presumably used by scouts and general managers when drafting players and those which predict actual NFL success are discussed.

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Effects of an individualized vs standardized Vitamin D supplementation on the 25(OH)D level in athletes. Nutrients, 15: 4747, 2023.

Abstract - Vitamin D is crucial to the health and performance of athletes. Although the exact vitamin D requirements for athletes have not been established, maintaining a 25(OH)D level of at least 40 ng/mL is considered beneficial. This randomized controlled intervention study aimed to determine whether an individual loading dose formula for vitamin D supplementation is more effective than standardized supplementation and suitable enough for athletes to meet a target value of 40 ng/mL. In a 10-week supplementation study conducted during the winter months in Germany, 90 athletes with insufficient vitamin D levels (25(OH)D < 30 ng/mL) were randomly assigned to receive either a universal dose of 2000 IU/day of vitamin D or a loading dose of 4000 IU/day, followed by a maintenance dose of 1000 IU/day. The total 25(OH)D concentration was measured from dried blood spots at three time points: at baseline, at the computed date of 40 ng/mL, and after the 10-week period. Additionally, a vitamin-D-specific questionnaire was issued. On the day when 25(OH)D blood concentrations of 40 ng/mL were calculated to prevail, the individualized group had a significantly higher 25(OH)D level than the standardized group (41.1 ± 10.9 ng/mL vs. 32.5 ± 6.4 ng/mL, $p < 0.001$). This study demonstrated that the examined formula is suitable enough for athletes to achieve a 25(OH)D concentration of 40 ng/mL. This indicates that a personalized approach is more effective than a one-size-fits-all approach in restoring adequate vitamin D levels in athletes.

Explaining variation in sweat sodium concentration: Effect of individual characteristics and exercise, environmental, and dietary factors. Journal of Applied Physiology, 133: 1250-1259, 2022.

Abstract - This study determined the relative importance of several individual characteristics and dietary, environmental, and exercise factors in determining sweat [Na^+] during exercise. Data from 1944 sweat tests were compiled for a retrospective analysis. Stepwise multiple regression ($P < 0.05$ threshold for inclusion) and T values were used to express the relative importance of each factor in a model. Three separate models were developed based on available independent variables: model 1 (1,944 sweat tests from 1,304 subjects); model 2 (subset with energy expenditure: 1,003 sweat tests from 607 subjects);

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model 3 (subset with energy expenditure, dietary sodium, and VO_{2max}: n = 48). Whole body sweat [Na+] was predicted from forearm sweat patches in models 1 and 2 and directly measured using whole body washdown in model 3. There were no significant effects of age group, race/ethnicity, relative humidity, exercise duration, pre-exercise urine specific gravity, exercise fluid balance, or dietary or exercise sodium intake on any model. Significant predictors in model 1 (adjusted $r^2 = 0.17$, $P < 0.001$) were season of the year (warm, T = -6.8), exercise mode (cycling, T = 6.8), sex (male, T = 4.9), whole body sweating rate (T = 4.5), and body mass (T = -3.0). Significant predictors in model 2 (adjusted $r^2 = 0.19$, $P < 0.001$) were season of the year (warm, T =-5.2), energy expenditure (T = 4.7), exercise mode (cycling, T = 3.6), air temperature (T = 3.0), and sex (male, T = 2.7). The only significant predictor in model 3 ($r^2 = 0.23$, $P < 0.001$) was energy expenditure (T = 3.8). In summary, the models accounted for 17%–23% of the variation in whole body sweat [Na^+] and energy expenditure and season of the year (proxy for heat acclimatization) were the most important factors.

Exploring the interplay between metabolic power and equivalent distance in training games and official matches in soccer: A machine learning approach. Frontiers in Physiology, 14:1230912, 2023.

Introduction: This study aimed to explore the interplay between metabolic power (MP) and equivalent distance (ED) and their respective roles in training games (TGs) and official soccer matches. Furthermore, the secondary objective was to investigate the connection between external training load (ETL), determined by the interplay of metabolic power and equivalent distance, and internal training load (ITL) assessed through HR-based methods, serving as a measure of criterion validity.

Methods: Twenty-one elite professional male soccer players participated in the study. Players were monitored during 11 months of full training and overall official matches. The study used a dataset of 4269 training games and 380 official matches split into training and test sets. In terms of machine learning methods, the study applied several techniques, including K-Nearest Neighbors, Decision Tree, Random Forest, and Support-Vector Machine classifiers. The dataset was divided into two subsets: a training set used for model training and a test set used for evaluation.

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Results: Based on metabolic power and equivalent distance, the study successfully employed four machine learning methods to accurately distinguish between the two types of soccer activities: TGs and official matches. The area under the curve (AUC) values ranged from 0.90 to 0.96, demonstrating high discriminatory power, with accuracy levels ranging from 0.89 to 0.98. Furthermore, the significant correlations observed between Edwards' training load (TL) and TL calculated from metabolic power metrics confirm the validity of these variables in assessing external training load in soccer. The correlation coefficients (r values) ranged from 0.59 to 0.87, all reaching statistical significance at $p < 0.001$.

Discussion: These results underscore the critical importance of investigating the interaction between metabolic power and equivalent distance in soccer. While the overall intensity may appear similar between TGs and official matches, it is evident that underlying factors contributing to this intensity differ significantly. This highlights the necessity for more comprehensive analyses of the specific elements influencing physical effort during these activities. By addressing this fundamental aspect, this study contributes valuable insights to the field of sports science, aiding in the development of tailored training programs and strategies that can optimize player performance and reduce the risk of injuries in elite soccer.

Explosive in not a term defined in the International System of Units and should not be used to describe neuromuscular performance. International Journal of Strength and Conditioning, Vol. 3(1): 1-10, 2023.

Abstract - Despite not being recognized by the International System of Units, explosive is a term often applied in sports science and professional practice. While associated with force, strength, power, performance, exercise, movements, contraction, action, and training, readers may be misled to believe that there are further analyses beyond power output, peak force, contractile rate of force development, and impulse. This critical review discusses the misuse of the term explosive in sports science literature, proposes alternatives, and encourages correct definitions of terms, units, and nomenclature to describe exercise performance. The suggestions provided in this review can help to reduce the confusion and perpetuation of an erroneous understanding of mechanical work, energy, and power in sports science.

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Fifth metatarsal strain distribution during cutting motions in soccer. Sports Biomechanics, Ahead of Print: 1-18, 2023.

Abstract - The objective of this study was to determine the fifth metatarsal strain generation mechanism during cutting motions performed while playing soccer using a finite element foot model. Five collegiate soccer players performed the side-step cutting and the cross-step cutting motions to measure the three-dimensional foot kinematics, ground reaction force, and plantar pressure distribution. In addition, a finite-element model of a foot consisting of bony structures, ligaments, and skin was constructed from computed tomography images. Simulations were conducted to perform the cutting motions, using the measured foot motion and distributed load on the plantar surface as boundary conditions for the model. During the side-step cutting, the maximum principal strain on the fifth metatarsal was correlated to forefoot adduction angle during stepping out. For cross-step cutting, the maximum principal strain was correlated with plantar pressure at the distal end of the fifth metatarsal. Therefore, to prevent a fracture, it is necessary to take measures to reduce the lateral bending deformation of the forefoot while stepping out during side-step cutting and to reduce the plantar pressure on the distal end of the fifth metatarsal during cross-step cutting.

From data to action: A scoping review of wearable technologies and biomechanical assessments informing injury prevention strategies in sport. BMC Sports Science, Medicine and Rehabilitation, 15: 169, 2023.

Background: The purpose of this scoping review was to evaluate the current use of technologies in sports settings for training adaptation and injury prevention. The review aimed to map the existing literature, identify key concepts and themes, and highlight gaps in research, thus offering guidance for future studies.

Methods: This study followed the guidelines of the PRISMA extension for scoping reviews and a search in four major databases was conducted.

Results: A total of 21 studies were included. The findings highlighted the widespread use of various technologies, including wearable devices and force plates, to monitor athletes' performance and inform

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evidence-based decision making in training and injury prevention. Variables such as Player Load, changes of direction, and acute chronic workload ratio were identified as key metrics in injury prediction.

Conclusions: This review uncovers a dynamic field of research in athlete injury prevention, emphasizing the extensive use of varied technologies. A key finding is the pivotal role of Player Load data, which offers nuanced insights for customizing training loads according to sport-specific demands, player positions, and the physical requirements of various activities. Additionally, the review sheds light on the utility of tools like force plates in assessing fatigue, aiding recovery, and steering injury rehabilitation, particularly in sports prone to knee and ankle injuries. These insights not only enhance our understanding of injury prevention but also provide a strategic direction for future research, aiming to boost athlete safety, performance, and career longevity.

Fundamentals or icing on top of the cake? A narrative review of recovery strategies and devices for athletes. Sports, 11: 213, 2023.

Abstract - The sport and athletic performance industry has seen a plethora of new recovery devices and technologies over recent years, and it has become somewhat difficult for athletes, coaches, and practitioners to navigate the efficacy of such devices or whether they are even required at all. With the increase in recovery devices and tools, it has also become commonplace for athletes to overlook more traditional, well-established recovery strategies. In this narrative review, we discuss recovery strategies in relation to the hierarchy of scientific evidence, classifying them based on the strength of the evidence, ranging from meta-analyses through to case studies and reports. We report that foam rolling, compression garments, cryotherapy, photobiomodulation, hydrotherapy, and active recovery have a high level of positive evidence for improved recovery outcomes, while sauna, recovery boots/sleeves, occlusion cuffs, and massage guns currently have a lower level of evidence and mixed results for their efficacy. Finally, we provide guidance for practitioners when deciding on recovery strategies to use with athletes during different phases of the season.

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Hamstring and knee injuries are associated with isometric hip and trunk muscle strength in elite Australian Rules and Rugby League players. Journal of Science and Medicine in Sport, Ahead of Print: 1-25, 2023.

Objectives: Smaller lumbar multifidus (LM) and larger quadratus lumborum (QL) cross sectional area (CSA) potentially increase lower limb injury risk. Including functional measures (force generating capacity) with structural (muscle CSA) may improve risk models. Relationships between isometric trunk and hip strength and morphology were explored with hamstring and knee ligament injury risk in Australian Football League (AFL) and National Rugby League (NRL) players.

Design: Prospective cohort study

Methods: During the AFL and NRL 2020 preseasons, isometric trunk and hip extensor strength (n = 153, 5 s (maximum strength) and 45 s (strength-endurance), and morphology (n = 238, LM L5 and QL CSA) were assessed. Logistic regression and decision trees explored associations between maximum strength, strength endurance, LM L5 and QL CSA, age, previous injury, and hamstring and knee ligament injuries.

Results: Hamstring (n = 19) and knee ligament (n = 13) injuries occurred. Greater strength endurance (OR = 0.42, 95%CI = 0.23-0.74, p = 0.004) and maximum strength (OR = 0.55, 95%CI = 0.31-0.94, p = 0.039) reduced hamstring injury risk. Knee ligament injury risk increased with larger LM L5 (OR = 1.66, 95%CI = 1.14-2.45, p = 0.008) and LM L5 to QL ratio (OR = 1.57, 95%CI = 1.13-2.23, p = 0.008). Decision tree models suggested low strength endurance (< 99Nm) was characteristic of hamstring strains yet those with high (≥ 1.33) LM L5 to QL ratio were protected. Knee ligament injuries were classified by larger (≥ 8.49 cm^2) LM L5 CSA, greater (≥ 1.25) LM L5 to QL ratio, and lower trunk and hip extensor maximum strength (< 9.24 N/kg).

Conclusions: Hamstring injury risk increased in players with lower trunk and hip extensor maximum strength maximum strength and strength-endurance. Knee ligament injury risk greater in players with larger LM L5 CSA and LM L5 to QL ratio and lower maximum trunk and hip extensor strength.

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Inertial measurement unit analysis for providing greater diagnostic value during the modified 5-0-5 change of direction test. The Journal of Sport and Exercise Science, Vol. 7(2): 54-61, 2023.

Abstract - Timing gates are currently the most common piece of equipment for measuring change of direction (COD) performance, however, they provide only a total time metric. A better understanding of the kinematics and kinetics during a COD movement beyond total time would provide coaches with a more comprehensive understanding of COD movement and how it can be improved. Therefore, the aim of this study was to determine the reliability of an inertial measurement unit (IMU) insole for measuring peak acceleration, peak deceleration, maximum speed, and ground contact time during a modified 5-0-5 change of direction (COD) test. Additionally, the strength of association between these IMU variables and timing light metrics was explored. Ten elite female netball athletes (age = 24.9 ± 5.0 years, height = 180.1 ± 6.5 cm, weight = 81.3 ± 15.0 kg) performed a modified 5-0-5 COD test across three testing occasions. Analysis revealed moderate to excellent relative consistency (ICC = 0.57 – 0.94) and acceptable absolute consistency (CV = 1.8 – 9.5%). Correlations ranged from 0.04 to 0.95, with peak acceleration having the strongest correlation with total time (r = 0.95). It appears that IMU insoles can be used to reliably measure performance during a COD task and provide additional diagnostics beyond time metrics.

Measurement properties of upper extremity physical performance tests in athletes: A systematic review. Brazilian Journal of Physical Therapy, Ahead of Print: 1-18, 2023.

Background: Upper extremity Physical Performance Tests (PPTs) have been used in sports contexts to provide functional status of the athletes. However, whether these tests present appropriate measurement properties to be considered a valuable measurement is not clear.

Objective: To systematically review the measurement properties of upper extremity PPTs in athletes.

Methods: Databases (e.g., Medline, EMBASE, CINAHL, SPORTDiscus, CENTRAL) were searched in March 2021. Two reviewers independently rated the methodological quality using the 4-point Consensus-based

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Standards for the Selection of Health Measurement Instruments (COSMIN) checklist. Quality of evidence was graded by measurement property for each test, considering the adequacy, the sample size, and the methodological quality of the studies.

Results: Fifteen studies were included with a pooled sample of 684 athletes. The PPTs analyzed were Arm-Jump Board Test, Closed Kinetic Chain Upper Extremity Stability Test (CKCUEST), Fin ger Hang Test, Medicine Ball Explosive Power Test, One-Arm Hop Test, Posterior Shoulder Endurance Test, Pull-Up Shoulder Endurance Test, Repetition to Failure Assessment, Seated Medicine Ball Throw Test (SMBT), Seated Single-Arm Shot-Put Test (SSPT), Shoulder Endurance Test, Two Arm Bent Hang Test, Unilateral Seated Shot-Put Test, and Upper Limb Rotation Test. Evidence synthesis provided moderate and high-quality evidence for sufficient inter-session and intra-session reliability of the CKCUEST, respectively. There was moderate evidence for sufficient inter-session reliability of the SSPT and for insufficient validity of the SMBT.

Conclusion: The CKCUEST and the SSPT are sufficiently reliable in athletes. More studies are needed to investigate other psychometric properties for these tests and other upper extremity PPTs.

Multiple ankle injuries are associated with an increased risk of subsequent concussion in National Football League players. Foot & Ankle Orthpaedics, Vol. 8(4): 1-6, 2023.

Background: There is a gap in the literature regarding musculoskeletal risk factors for concussion within the National Football League (NFL), which is an avenue that must be explored to promote player safety given the high incidence of both injury types. This study aims to observe if ankle injuries are associated with an increased risk of subsequent concussion in NFL players.

Methods: The public online database ProFootballReference.com was used to identify ankle injuries and concussions in NFL players from the 2009-2010 to 2019-2020 seasons. Multivariable logistic regression for subsequent concussion and ankle injury was performed, adjusting for body mass index (BMI), age, and player position. For descriptive statistics, unpaired t tests with unequal variance were performed for

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continuous variables, including BMI and age. χ2 testing was performed for categorical variables, including player position, and whether the position was offensive, defensive, or on special teams.

Results: Of the 5538 NFL players included in the study, 941 had an ankle injury, 633 had a concussion, and 240 had both an ankle injury and a subsequent concussion. The adjusted odds ratio (aOR) for concussion following a single ankle injury was 0.90 (95% CI 0.72-1.14, P=.387); however, the aOR for concussion following multiple ankle injuries was 2.87 (95% CI 1.23-6.75, P=.015). Special teams players had the lowest risk for concussion (aOR 0.17, 95% CI 0.069-0.36, P<.001) following ankle injury, and there was no significant difference in risk between offense and defense (aOR 0.91, 95% CI 0.77-1.08, P=.295).

Conclusion: Multiple ankle injuries were associated with an increased risk of a subsequent concussion after adjusting for BMI; player position; and offense, defense, or special teams designation. These findings can inform injury prevention practices in the National Football League.

Napping behavior, daytime sleepiness, and arousal in high performance athletes and non-athlete controls. Journal of Sports Sciences, Ahead of Print: 1-9, 2023.

Abstract - Napping offers a strategy to manage sleep, aid recovery and enhance performance in elite sport. However, relatively little research attention has focused on the natural history of athlete napping or tested the widely held assumption that athlete napping is mainly a consequence of degraded night-time sleep. Within a sample of 158 team (n = 76) and individual (n = 80) sport athletes, and 82 non-athlete controls, we analyzed napping behavior in relation to sleep quality, daytime sleepiness, and pre-sleep somatic and cognitive arousal. There was no significant association between athlete/non-athlete status and the prevalence, frequency, or duration of naps. Comparisons of athlete nappers and non-nappers found no significant differences in sleep quantity or quality. While nap propensity was significantly related to higher daytime sleepiness, this influence was moderated or augmented by levels of pre sleep cognitive arousal. For some nappers, those with higher levels of arousal may need to be sleepier than those with lower levels of arousal in order to successfully initiate daytime sleep. Approximately 50% of athletes did not nap. If the benefits of athlete napping are to be fully exploited,

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the needs of this substantial group for whom napping may be problematic should be recognized and addressed.

Nighttime smartphone use, sleep quality, and mental health: Investigating a complex relationship. Sleep, 46: 1-12, 2023.

Study Objectives: This study investigated the complex relationship between nighttime smartphone use, sleep, and mental health among adult populations in Denmark.

Methods: Data from three interconnected samples (aged 16–89 years) from the SmartSleep Study included 5798 individuals with sur vey and register data; 4239 individuals also provided high-resolution smartphone tracking data. Logistic regression models and causal discovery algorithms, which suggest possible causal pathways consistent with the underlying data structure, were used to infer the relationship between self-reported and tracked nighttime smartphone use, self-reported sleep quality, mental health indicators, and register-based psychotropic medication use.

Results: Frequent self-reported nighttime smartphone use was associated with high perceived stress (OR: 2.24, 95% CI = 1.42 to 3.55) and severe depressive symptoms (OR: 2.96, 95% CI = 2.04 to 4.28). We found no clear associations between tracked nighttime smartphone use and mental health outcomes, except for the cluster that used their smartphones repeatedly during the sleep period, which was associated with severe depressive symptoms (OR = 1.69, 95% CI = 1.24 to 2.31). Poor sleep quality (vs. good sleep quality) was associated with high perceived stress (OR = 5.07, 95% CI = 3.72 to 6.90), severe depressive symptoms (OR = 9.67, 95% CI = 7.09 to 13.19), and psychotropic medication use (OR = 2.13, 95% CI = 1.36 to 3.35). The causal discovery models suggest that nighttime smartphone use affects mental health through both problematic smartphone use and poor sleep quality.

Conclusion: The complex relationship between nighttime smartphone use, sleep, and poor mental health may create a vicious circle over time, and nighttime smartphone use may constitute a potential leverage point for public health interventions aimed at improving sleep and mental health.

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Open fractures in National Football League athletes: Analyzing performance and return to sport. Spartan Medical Research Journal, 8(1):87846, 2023.

Introduction: Open fractures are potentially devastating injuries for the professional athlete. We sought to compare return to sports (RTS) and performance in National Football League (NFL) athletes sustaining open versus closed fractures.

Methods: NFL athletes with surgically treated open and closed fractures of the forearm, tibial shaft, and ankle from 2009-2018 were identified through publicly available reports and records. Data including demographics, RTS, career duration, and the approximate value performance metric before and after injury were collected. Statistical analyses were performed comparing open to closed injuries. Continuous variables were compared using Mann-Whitney U or two sample t- tests while categorical variables were compared using Fisher's exact test.

Results: Ninety-five athletes met inclusion criteria (10 open and 85 closed fractures). Overall, 90% (n = 9) returned to sport after an open injury and 83.5% (n = 71) returned after closed injury with a median time missed of 48.9 (range 35.1 - 117.4) weeks and 43.0 (range 2.4 - 108.0) weeks, respectively. Athletes undergoing forearm surgery were able to return sooner, at around 20.8 weeks, and ankle fractures conferred the lowest return rate at 80% (n = 48). There were no significant differences in career duration and post-injury performance between open or closed fracture cohorts.

Conclusions: Although open fractures are relatively uncommon injuries seen in NFL athletes, our study suggests RTS for these players is high. Athletes undergoing surgical treatment for open fractures had similar RTS rates, performance metrics, and career durations compared to those with comparable closed fractures. This information can provide guidance for providers counseling elite athletes on postoperative expectations.

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Potential health risks of foodborne performance-enhancing drugs in competitive sports. Heliyon, 9:321104, 2023.

Abstract - Athletes need to consume a significant amount of energy during prolonged training and in high-intensity competition. It is necessary for them to take nutritional foods that recharge their bodies. However, in sporting events of recent years, both domestic and international, many positive drug tests are found to be caused by the ingestion of foods that contain performance-enhancing drugs (PEDs). As a result, the prevention and control of PEDs in food supply have drawn increasing attention. For better prevention and control, the first step is to understand the food contaminants – PEDs. This study has categorized PEDs through their presence in animal-derived foods, plant derived foods, and synthetic nutritional supplements in competitive sports. It investigates the potential risks of foodborne doping using techniques such as external addition and endogenous component analysis. This research explored the causes of PEDs in food and their negative effects on athletes and proposed measures to ensure the safety of nutritional substances in competitive sports. PEDs in animal-derived foods include β-adrenergic agonists, anabolic steroids, and glucocorticoids, which can be found in meat and ox penis, amongst other food sources. In contrast, PEDs in plant-derived foods include alkaloids, higenamine, and zeranol, which can be found in coffee, tea, Sichuan pepper, custard apple, and cereal. Performance-enhancing drugs (PEDs) that are often added to synthetic supplements include creatine, traditional Chinese herbs, 1, 3-dimethylbutylamine (DMAA), sibutramine, ephedrine, and methylhexanamine. Targeted anti-doping training should be provided to athletes. In addition, the latest domestic and international standards and regulations regarding PEDs or prohibited and restricted ingredients in foods should be tracked in real-time. The control list for performance-enhancing drugs in food should be continually updated and refined. Research on detection methods for performance-enhancing drugs in food should also be advanced. Moreover, market surveillance and law enforcement should be strengthened to ensure that sports foods meet national safety standards before they enter the market. This paper provides workable solutions to clarify the types and scope of performance-enhancing drugs in food, aiming to improve the prevention and control of PEDs in animal-derived foods, plant-derived foods, and supplements in major sporting events.

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Quantifying demands on the hamstrings during high-speed running: A systematic review and meta-analysis. Scandinavian Journal of Medicine and Science in Sports, Ahead of Print: 1-21, 2023.

Introduction: Hamstring strain injury (HSI) remains a performance, economic, and player availability burden in sport. High-speed running (HSR) is cited as a common mechanism for HSI. While evidence exists regarding the high physical demands on the hamstring muscles in HSR, meta-analytical synthesis of related activation and kinetic variables is lacking.

Methods: A systematic search of Medline, Embase, Scopus, CINAHL, SportDiscus, and Cochrane library databases was conducted in accordance with the PRISMA 2020 guidelines. Studies reporting hamstring activation (electromyographic [EMG]) or hamstring muscle/related joint kinetics were included where healthy adult participants ran at or beyond 60% of maximum speed (activation studies) or 4m per second (m/s) (kinetic studies).

Results: A total of 96 studies met the inclusion criteria. Run intensities were categorized as “slow,” “moderate,” or “fast” in both activation and kinetic based studies with appropriate relative, and raw measures, respectively. Meta-analysis revealed pooled mean lateral hamstring muscle activation levels of 108.1% (95% CI: 84.4%–131.7%) of maximal voluntary isometric contraction (MVIC) during “fast” running. Meta-analysis found swing phase peak knee flexion internal mo ment and power at 2.2 Newton meters/kilogram (Nm/kg) (95% CI: 1.9–2.5) and 40.3 Watts/kilogram (W/kg) (95% CI: 31.4–49.2), respectively. Hip extension peak moment and power was estimated as 4.8Nm/kg (95% CI: 3.9–5.7) and 33.1W/kg (95% CI: 17.4–48.9), respectively.

Conclusions: As run intensity/speed increases, so do the activation and kinetic demands on the hamstrings. The presented data will enable clinicians to incorporate more objective measures into the design of injury prevention and return-to play decision-making strategies.

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Review of media representation of sport concussion and implications for youth sports. Clinics in Sports Medicine, Ahead of Print: 1-14, 2023.

KEY POINTS

- Many individuals learn about sport-related concussion (SRC) through various forms of media (newspaper, online news, social media, film, and so forth)
- Popular media does not always portray SRC accurately and there are many examples of misinformation in reporting.
- Knowledge about SRC varies considerably based on several factors, but accurate understanding of the injury does not always correlate to appropriate action.
- Given the high utility of various forms of media usage, this presents an opportunity to convey accurate SRC information.

Introduction - Participation in youth sport confers numerous mental health and physical benefits.1 Nonetheless, participating in physical activity comes with inherent risks. Sport-related concussion (SRC), a type of mild traumatic brain injury (mTBI), is a risk associated with any physical activity. The impact of SRC and the potential long-term neurocognitive effects have been a mainstay in media over the past 2 decades. This has been driven by cases in which the postretirement declines of high-profile athletes have been attributed to SRC and repetitive brain impacts sustained during playing careers. Press coverage of these events has likely had a significant impact on public opinion regarding SRC and participation in youth sport. Rates of participation in some high school (HS) sports such as football and basketball have decreased in recent years for the first time in decades, notwithstanding the COVID-19 pandemic. Athlete and parent concern about SRC is frequently cited as a driving factor, and for those who do remain engaged in sport, fear of SRC remains high. Other beliefs about the short- and long-term impact of SRC are not always consistent with up-to-date sci ence.2 As varying forms of media remain major sources of SRC-related information, an exploration of its impact is worthwhile. Here, the authors examine some of the ways that media has represented SRC and how this representation may continue to influence youth sport participation today.

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Sciatic nerve compression after a chronic proximal hamstring tear: A report of two cases and a narrative review of the literature. Life, 13:1762, 2023.

Abstract - Proximal hamstring tears are among the most common injuries afflicting athletes and middle-aged individuals. Sciatic nerve compression after a proximal hamstring injury, which can occur due to scar formation and subsequent irritation or compression of the nerve, is an infrequent but severe complication with few cases documented in the literature. No evidence is available about the optimal treatment for sciatic nerve symptoms after proximal hamstring injuries. In this case report, we present two cases involving patients primarily treated conservatively at another institution after suffering from a proximal hamstring injury and developing sciatic nerve symptoms over the course of a few months. Both were treated with open neurolysis at our institution without reattachment of the ruptured muscles to the ischial tuberosity due to the chronicity of the injuries. Both patients exhibited neurological symptoms over two years, which recovered after surgery. These two cases show that neurolysis of the sciatic nerve without reattachment of the proximal hamstring muscles is an applicable option for the treatment of chronic proximal hamstring tears with sciatic nerve compression. Further studies will be needed to validate this hypothesis.

The effects of knee ligament load using simulated hip abductor and hamstring muscle strengthening during cutting maneuver. Medicine, 102:46, 2023.

Background: This study aimed to analyze knee ligament of load and joint moment to simulate the strengthening of the hip abductor and hamstring muscles using musculoskeletal modeling, thereby contributing to decrease of knee ligament load.

Methods: Forty participants (age: 21.85±1.90 years; height: 1.76±0.06 m; body mass: 68.5±7.06kg) were instructed to run at 4.5±0.2 m/s from a 5 m distance posterior to the force plate, land their feet on the force plate, and perform the cutting maneuver on the left. In the musculoskeletal modeling, the hip abductor and hamstring muscles were targeted to construct a model with a 30% increase in the contraction force of the hip abductor, hamstring, and both 2 muscles. The variables were the ligament force and knee joint moment. One-way repeated measure ANOVA and Bonferroni test were used to compare the abductor/hamstring, abductor, hamstring and control models.

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Results: There were significant differences in anterior bundle of the anterior cruciate ligament (ACL) (P = .001), inferior bundle of the superficial layer of the medial collateral ligament (MCL) (P = .016), and posterior bundle of the superficial layer of the MCL (P = .022) force. The post hoc showed that the hamstring model had lower anterior bundle of the ACL and inferior bundle of the superficial layer of the MCL than the abductor/hamstring and abductor models (P < .05) and lower posterior bundle of the superficial layer of the MCL than the abductor and control models (P < .05). There was a significant difference in the adduction (P= .028) and internal rotation moments (P = .014). The post hoc showed that both moments were lower in the hamstring model than in the other models (P < .05).

Conclusions: The hamstring strengthening may contribute significantly to preventing ACL or MCL injury by reducing knee ligament load.

The effects of relaxation techniques following acute, high intensity football training on parasympathetic reactivation. Frontiers in Sports and Active Living, 5:1267631, 2023.

Background: Evidence shows relaxation techniques reactivate the parasympathetic nervous system (PNS) following physiological stressors such as exercise. As such, these techniques may be useful following exercise training of high intensity sports, like collegiate football.

Purpose: To evaluate the impact of mindfulness and rest activities on PNS reactivation following training sessions, in a sample of Division-I collegiate, male football athletes.

Methods: This study employed a cross-sectional, pre-post experimental design among 38 football athletes. Following three training sessions, each separated by one week, athletes were exposed to three groups: mindfulness, rest, and no-intervention. Athletes in the mindfulness group laid supine in a darkened room, while performing 15 min of guided breathing and body scans. The rest group remained seated in a lighted room, performing 15 min of restful activities (e.g., talking). The no-intervention group was instructed to perform usual post-training activities (e.g., showering). Heart rate (HR), respiration rate (RR) and two HR variability (HRV) indices were measured via an armband monitor (Warfighter Monitor, Tiger Tech Solutions, Inc, Miami, FL) equipped with electrocardiographic and

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photoplethysmography capabilities. HRV indices included standard deviation of the N-N intervals (SDNN) and root mean square of successive RR interval differences (rMSSD). Within and between-group differences were determined via analysis of variance (ANOVA) and corrected for multiple comparisons familywise error.

Results: Statistically significant reductions in HR and RR were observed across all groups: −81.6, −66.4, −40.9 bpm and −31.7, −26.9, and −19.0 breaths·min−1, respectively. The mindfulness and rest groups exhibited a larger within-group reduction in HR and RR compared to the no-intervention group, $p < 0.0000$. Additionally, the mindfulness group showed a larger reduction in HR and RR compared to the rest group, $p < 0.05$. Post-intervention HR and RRs were significantly lower in the mindfulness group relative to the no-intervention group (77.0 vs. 120.1 bpm, respectively). Similar results were observed for RR (15.0 vs. 23.6 breaths·min−1, respectively) and HRV indices (SDNN: 46.9 vs.33.1 ms and rMSSD: 17.9 vs. 13.8 ms, respectively) Athletes in the rest group showed significantly lower post-intervention HR (−30.2 bpm, 89.9 vs. 120.1 bpm, respectively), RR (−4.3 breaths·min−1, 19.3 vs. 23.6 breaths·min−1, respectively) and significantly higher HRV (SDNN: 42.9 vs. 33.1 ms and rMSSD: 16.7 vs.13.8 ms, respectively) compared to their no-intervention counterparts.

Conclusions: Our findings suggest that athletes engaging in either 15-minute guided mindfulness or rest activities (e.g., sitting) post training, may facilitate PNS reactivation. Implementing these strategies may accelerate recovery, improving performance. Longitudinal, randomized controlled trials among diverse sports are encouraged.

The Molecular Athlete: Exercise physiology from mechanisms to medals. Physiological Reviews, 103: 1693-1787, 2023.

Abstract - Human skeletal muscle demonstrates remarkable plasticity, adapting to numerous external stimuli including the habitual level of contractile loading. Accordingly, muscle function and exercise capacity encompass a broad spectrum, from inactive individuals with low levels of endurance and strength to elite athletes who produce prodigious performances underpinned by pleiotropic training-induced muscular adaptations. Our current understanding of the signal integration, interpretation, and output coordination of the cellular and molecular mechanisms that govern muscle plasticity across this

continuum is incomplete. As such, training methods and their application to elite athletes largely rely on a “trial-and-error” approach, with the experience and practices of successful coaches and athletes often providing the bases for “post hoc” scientific enquiry and research. This review provides a synopsis of the morphological and functional changes along with the molecular mechanisms underlying exercise adaptation to endurance- and resistance-based training. These traits are placed in the context of innate genetic and interindividual differences in exercise capacity and performance, with special consideration given to aging athletes. Collectively, we provide a comprehensive overview of skeletal muscle plasticity in response to different modes of exercise and how such adaptations translate from “molecules to medals.”

To train or not to train (on match day)? Influence of a priming session on match performance in competitive elite-level soccer. Journal of Sports Sciences, Ahead of Print: 1-9, 2023.

Abstract - This study examined the effect of a morning priming session consisted of stretching, mobility, core and lower-body resistance, and reactive agility exercises on match performance in elite-level soccer players. Information on physical and technical performance was collected during competitive matches (n = 32) and subsequently classified as those preceded or not preceded by a priming session. Results showed significantly increased (i) distance covered overall (Cohen’s d (d) = 0.34, p = 0.011), in moderate-intensity (d = 0.52, p = 0.001) and high-intensity running (d = 0.30, p = 0.024), and (ii) frequency of duels (d = 0.26, p = 0.050) in matches preceded by a priming session compared to those not preceded by a priming session. These findings show that players’ physical performance was notably enhanced due to the pre match routines that the team implemented on match day, while technical performance was in general similar irrespective of whether a priming session was implemented or not before the match. Therefore, morning priming session implemented on match day can be recommended as an effective pre-match strategy to increase players’ physical performance in soccer without negative consequences on technical performance.

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Pro Football Sports Science Research Update

Turf toe injuries in the athlete: An updated review of treatment options, rehabilitation protocols, and return to play outcomes. Current Reviews in Musculoskeletal Medicine, Ahead of Print: 1-12, 2023.

Purpose of Review: First metatarsophalangeal joint sprains or turf toe (TT) injuries occur secondary to forceful hyperextension of the great toe. TT injuries are common among athletes, especially those participating in football, soccer, basketball, dancing, and wrestling. This review summarizes the current treatment modalities, rehabilitation protocols, and return-to-play criteria, as well as performance outcomes of patients who have sustained TT injuries.

Recent Findings: Less than 2% of TT injuries require surgery, but those that do are typically grade III injuries with damage to the MTP joint, evidence of bony injury, or severe instability. Rehabilitation protocols following non-operative management consist of 3 phases lasting up to 10 weeks, whereas protocols following operative management consist of 4 phases lasting up 20 weeks. Athletes with low-grade injuries typically achieve their prior level of performance. However, among athletes with higher grade injuries, treated both non-operatively and operatively, about 70% are expected to maintain their level of performance.

Summary: The treatment protocol, return-to-play criteria, and overall performance outcomes for TT injuries depend on the severity and classification of the initial sprain. For grade I injuries, players may return to play once they experience minimal to no pain with normal weightbearing, traditionally after 3–5 days. For grade II injuries, or partial tears, players typically lose 2–4 weeks of play and may need additional support with taping when returning to play. For grade III injuries, or complete disruption of the plantar plate, athletes lose 4–6 weeks or more depending upon treatment strategy.

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Pro Football Sports Science Research Update

Where to next for hamstrings? A biomechanical and anatomical perspective. British Journal of Sports Medicine, Ahead of Print: 1-2, 2023.

In recent years, there has been rising interest and debate on the importance of addressing running mechanics for preventing running-related hamstring injury. It is a logical and intriguing avenue of research given the high prevalence of such injuries. In theory, lower-limb mechanics directly relate to hamstring musculotendinous demands when running—the relative positions of the hip and knee joints, and pelvic tilt, will deter mine the hamstring muscle–tendon unit (MTU) lengths. Notably, hamstring MTUs during running are subject to large strain and force during late swing. However, there is a paucity of research prospectively investigating the relationship between running mechanics, or hamstring MTU dynamics, and injury, likely because it is challenging to prospectively obtain large samples of data from injured athletes.

ASSESSING RUNNING MECHANICS

To better understand the relationship between running mechanics and hamstring injury, it can be useful to examine multiple case studies (or case series), the advantage being they offer constructive observation(s) towards novel concepts in ecologically valid contexts. Schuermans et al found that four injured soccer players displayed greater anterior pelvic tilt during mid-swing and greater thoracic lateral flexion during the late swing phase of sprinting. Similarly, in a study of three rugby players2 who subsequently sustained hamstring injuries, a tendency for greater thoracic lateral flexion during the late swing phase as well as greater knee power absorption and hip extension moments was observed. As noted earlier, given the attachment sites of the hamstring muscles, it is logical that the mechanics of the hip, knee and pelvis may relate to hamstring strain and injury risk. It is also likely that the observed mechanics are inter-related, as any increase in joint moment magnitude that increases the acceleration of a segment (eg, a larger hip extension moment increasing the acceleration of the thigh) may influence adjacent joints and other segments through intersegmental torques. This may be especially evident during the swing phase where muscle generated torques are primarily opposing motion-dependent torques. The influence of thoracic lateral flexion is perhaps less clear, but it has been postulated that this may relate to insufficient dissociative capacity within the lumbo–pelvic– hip complex. Increased thoracic lateral flexion may also relate to the observed hip extension moment in injured athletes, given that during late swing, energy generated by the hip is transferred proximally to the trunk. Collectively,

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these data from soccer and rugby show emerging evidence that running mechanics may influence hamstring injury risk. However, data are from a small number of case studies; therefore, although insightful, evidence remains inconclusive. Further, both studies examined mechanics in a laboratory setting without the presence of fatigue. This is important to note given that many injuries occur in the latter half of game periods (eg, halves or quarters) under fatigued conditions, which may alter running mechanics. The relationship between fatigue-affected running mechanics and hamstring injury warrants further research. Nevertheless, these findings prompt us to consider how mechanics at the musculotendon level may influence injury susceptibility.

MUSCULOTENDON MECHANICS

Several studies have attempted to link hamstring MTU mechanics to injury. Hamstring MTU lengths and forces directly prior to injury have been determined from serendipitous measurement during running-based injury. These studies showed that during late swing, the injured leg's biceps femoris long head (BFlh) reached a greater length and produced greater net hamstring MTU force compared with the uninjured leg. These data suggest that BFlh strain and force are key mechanisms for injury. However, the athlete(s) in those studies had sustained recent recurrent hamstring strains on the injured leg prior to testing; therefore, it may be argued the observed MTU dynamics may be a consequence of the prior injury. A more recent report from two different injury cases provides additional evidence linking hamstring MTU dynamics to injury. In this study, data from the pre-season prior to injury showed that the subsequently injured BFlh of both athletes displayed the greatest strain of all hamstring MTUs during late swing. A strength of this study was that MTU force was normalized to physiological cross-sectional area (PCSA), as PCSA reflects the force-generating capacity of the muscle. It was seen that BFlh MTU force was equal to, or greater than, semimembranosus force during late swing. This suggests that the BFlh muscle was under considerable stress during late swing due to the large force in a muscle with a relatively small PCSA. To our knowledge, hamstring MTU force has not previously been considered in this way; however, this insight prompts us to contemplate how information about running mechanics and MTU dynamics could be enhanced with anatomical information.

HAMSTRING SHAPE

When considering anatomical information alongside running mechanics and MTU dynamics, it is important to have some appreciation of what anatomical research has been conducted. Early research

focused largely on cadaveric measures of overall morphology such as PCSA, volume and MTU length. Since then, research has explored the importance of muscle architecture, in particular fascicle length. This has had considerable practical implications for injury prevention programs, reinforcing the importance of eccentric hamstring exercises or exercises that stress the muscle at long lengths, to elicit fascicle elongation. More recently, advanced methods for capturing information about muscle morphology, including complex three-dimensional shape, have been used. These methods have the potential to provide much more information about hamstring morphology: how it varies between groups and how it may relate to injury risk. Sutherland et al applied statistical shape modelling techniques to discriminate hamstring muscle shapes between rugby and sprinting athletes. Three-dimensional shape features such as size, curvature and torsion were found to distinguish the cohorts. This suggests that meaningful variation in hamstring shape exists between individuals and populations and can be captured in a way that could be used for understanding injury risk. To date, no studies have investigated whether three-dimensional muscle shape may relate to injury risk. However, this is an interesting avenue for future research, given the theoretical link between muscle shape and function. Variations in shape theoretically reflect a variation in the muscle's line of action, which could influence MTU forces and forces transferred to the bone to produce joint movement. Given that biomechanical research has suggested that MTU forces may influence injury susceptibility, it is plausible that muscle shape could provide additional insight into hamstring muscle function and could be linked to injury. Further, information about muscle shape could be used within musculoskeletal models to further integrate analyses of biomechanics and anatomy in relation to injury. Moving forward, to gain a better understanding of hamstring function and mechanisms for injury, we should consider joint mechanics along with hamstring MTU dynamics and anatomy rather than exploring each in isolation.

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